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Selecting active frames for action recognition with vote fusion method

Hoang Tieu Binh
Hanoi National University of Education
Hanoi, Vietnam
Email: htbinh@hnue.edu.vn

Ma Thi Chau
University of Engineering and Technology
Hanoi, Vietnam
Email: chaumt@vnu.edu.vn

Akihiro Sugimoto
National Institute of Informatics
Tokyo, Japan
Email: sugimoto@nii.ac.jp

Bui The Duy
University of Engineering and Technology
Hanoi, Vietnam
Email: duybt@vnu.edu.vn

ABSTRACT

Recent applications of Convolutional Neural Networks, especially 3-Dimensional Convolutional Neural Networks (3DCNNs) for human action recognition (HAR) in videos have widely used. In this paper, we use a multi-stream framework which is a combination of separated networks with different kinds of input generated from a unique video dataset. We study various methods for extracting best frames in videos for action representation and find the way to fuse multiple networks for better recognition. We make the following work: first, we propose a method to extract the active frames (called Selected Active Frames - SAF) from videos to build datasets for 3DCNNs in video classification problem. Secondly, we propose a mixed fusing approach called Vote fusion which is considered as an effective fusion method for ensembling multi-stream networks. We evaluate the proposed approach to solving action recognition problem. We carry out this approach on three well-known datasets (KTH, HMDB51, and UCF101). The results are also compared to the state-of-the-art results to illustrate the efficiency and effectiveness in our approach.

Keywords: Action recognition, fusion method, 3D convolutional neural networks, multimodal network.

I. INTRODUCTION

Video-based human action recognition has become an attractive topic in recent years. Since Alex Krizhevsky *et al.* showed the best result in Imagenet 2012 competition [1], there is a revolution on action recognition matter. Going with stronger hardware infrastructure, larger action database, the recognition results increase day by day. From the first stage of the recognition process, data were usually used for extracting features before applying some traditional recognizing procedures. The methods SIFT (Scale Invariant Feature Transform) [2], SURF (Speeded Up Robust Feature) [3] and HOG (Histogram of Oriented Gradients) [4] are the examples.

Nowadays, deep learning shows some huge advantages among other feature representation methods. Especially Convolutional Neural Network (CNN) is one of the deep structures which achieves state-of-the-art results in various fields. In the domain of image classification, Convolutional Networks (ConvNets) [5] demonstrates the powerful potential of learning visual representations [1] from raw visual data.

However, the expansion of CNNs to action recognition in videos in recent research remains has not been able to achieve significant improvements over traditional hand-craft features for video-based action recognition. Beside this, 2D-CNN architectures or any other previous methods are not able to exploit the important characteristics of videos like cube information which describes motion and action in videos [6].

Over the last five years, 3DCNNs and its variations for action recognition often focus on short video intervals with various number of frames from 1 to 16 [7], [8], [9], [10],

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[11], [12]. Particularly, Varol *et al.* used 100-frames dataset as temporal resolutions for spatio-temporal convolutional networks [13].

Equipped with large-scale training datasets like Imagenet [14], MS-Celeb-1M [15], THUMOS Dataset [16], and Activitynet [17]. Deep convolutional networks are the most chosen method for the still-image recognition tasks such as face, object, scene, human action recognition [18], [19], [20], [21].

In this paper, we have two contributions related to action recognition in videos. First, we propose a method called Active Selected Frame for extracting a dataset from videos for the training process. Secondly, we present a novel fusion method for ensembling multiple networks. The explanation of these two contributions is described in Section III and Section IV. The result is evaluated and commented in Section VII.

II. Related work

Several recent works on how to extract motion information and deploy it in the 3DCNNs show the significant improvement results. The nature of using 3DCNN in modeling actions in a temporal manner is no argument. From the successful C3D network implemented by Du Tran et al [9], there are many versions of C3D which deployed end-to-end networks which trained directly from videos. [7] used two-stream convolutional networks for action recognition in videos which split data into two datasets called spatial stream and temporal stream. The first network uses a single frame as input data, the second network use multi-frame optical flow (OF) as temporal information. Both network's outputs will be fused by a class score ensemble to generate classification at the end. Varol *et al.* in [13] used Long-term Temporal Convolutions (LTC) to learn video representations to demonstrate LTC-CNN with increased temporal extents will improve the accuracy of action recognition. Hakan Bilen *et al.* introduced dynamic images (DI) which are extracted from videos with a weighted number applying on each frame [10]. In [22], to encode the information extracted from video, Optical Flow Co-occurrence Matrices (OFCM) which based on the co-occurrence matrices, computed over the optical flow field are captured.

Also related to our work is the fusion methods which show some optimistic signal when applying to multiple networks. By using multiple streams neural network with various fusing methods could issue better results [23]. [6] uses two-stream network fusion for video action recognition on both spatial and temporal information. Boosting fusion method is used in [24] demonstrated that the accuracy improves 7.2% and 7.9% when executing on UCF101 [25] and HMDB51 [26].

III. Fusion methods

Fusion method is the combination of the two or multiple layers from two or multiple networks. They are powerful procedures which improve networks' performance. It reduces the variance portion in the bias-variance decomposition of the prediction error. There is a number of ways of fusing multiple networks. We have experimented with different fusion methods that all tend to contribute to accuracy improvement. In addition, the tradeoff between a number of models and their complexity has been investigated and we show that fusing learning may lead to accuracy gains along with a reduction in training time. Concretely, we go further with three popular fusion methods which mentioned by Christoph Feichtenhofer in [6].

Without loss of generality, assume that we have two feature maps $x^a \in \mathbb{R}^{H\times W\times D}$ and $x^b \in \mathbb{R}^{H\times W\times D}$, we define a function $f : x^a, x^b \to y$ which fuses these map a, b to produce $y \in \mathbb{R}^{H\times W\times D}$ where H, W, D as the height, width, depth of channels of the respective feature maps. When applying to convolutional neural network architectures, consist of convolutional, pooling, fully-connected layers, there are multiple options for applying f at different layers, for example, early-fusion, late-fusion or multiple layer fusion [8].

Average fusion: $y^{sum} = f^{avg}(x^a, x^b)$ evaluates the average of the two feature maps at the same location i, j and feature channel d:

$$y^{avg}_{i,j,d} = avg(x^a_{i,j,d}, x^b_{i,j,d}), \tag{1}$$

where $1 \leqslant i \leqslant H, 1 \leqslant j \leqslant W, 1 \leqslant d \leqslant D$ and $x^a, x^b, y^{sum} \in \mathbb{R}^{H\times W\times D}$.

Max fusion: Similar to Average fusion in equation (1), $y^{max} = f^{max}(x^a, y^b)$ takes the maximum of the two feature maps:

$$y^{max}_{i,j,d} = max(x^a_{i,j,d}, x^b_{i,j,d}), \tag{2}$$

where i, j, d are defined as (1), $y^{max} \in \mathbb{R}^{H\times W\times D}$.

Concatenation fusion: $y^{cat} = f^{cat}(x^a, x^b)$ combine two feature maps at the same location i, j across the feature channel d.

$$\begin{aligned} y^{cat}_{i,j,2d} &= x^a_{i,j,d}, \\ y^{cat}_{i,j,2d-1} &= x^b_{i,j,d}, \end{aligned} \tag{3}$$

where $y^{cat} \in \mathbb{R}^{H\times W\times 2D}$.

Vote fusion: Based on the classification results of all networks, the labels which most networks recognize are chosen. If all networks have the resembled results, max/average fusion on these networks are applied to get the final recognition results (noted that, we compare the classification results on each test sample). Comparing to Majority Voting [27], Vote fusion method is more flexible because the classification does not only depends on the voting but also uses fusion scores of all networks. We carry out experiments on multi-model frameworks which include more than two networks. The special situation for this method in the circumstance that we have only one or two networks, in this case, we use Average fusion or Max fusion method as mentioned above. The Table III provide the comparison between two methods.

IV. Extracting data method

In a video, there is a series of frames which contain similar information. Our work is looking for a way to select the most informative frames which represent for the whole clip.

Assume that the difference between two consecutive frames is small, we recognize that the more the differences between the two frames, the more motions are expressed. In our work, to measure the difference between two consecutive frames, we calculate the Euclidean distance [28] between them (the distance between their corresponding points in the image space, for all of channels).

The Euclidean distance of two images x, y of fixed size M by N is written by:

$$d_E^2(x, y) = \sum_{i,j=1}^{MN} g_{i,j}(x^i - y^i)(x^j - y^j) = (x - y)^T G(x - y). \tag{4}$$

Where the symmetric matrix $G = (g_{i,j})_{MN \times MN}$ will be referred to as metric matrix.

The most informative frame is the frame which has the largest distance between two continuous images. We called it Selected Active Frame (SAF) and it is calculated by using the following procedure:

Algorithm 1: Extract frame by finding the largest distance between adjacent images in a segment of a video clip

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Input : One segment with fixed number of frames
Output: The most active frame

```
d_max = 0
for i ← 1 to number of frames do
    F_i ← the i^th frame
    F_{i+1} ← the next frame after F_i
    d_i = Euclidean_distance(F_{i+1}, F_i)
    if d_i > d_max then
        d_max = d_i
        F_max = F_i
    end
end
return F_max
```

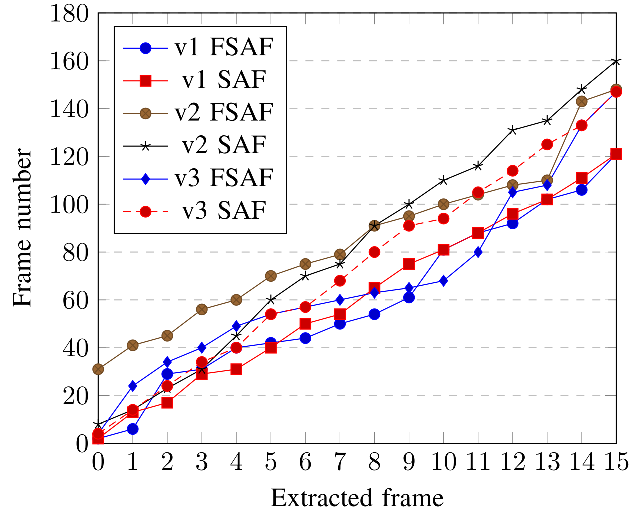

Fig. 1: Comparing between FASF and SAF method, v1, v2, v3 denote for video 1, video 2, video 3, respectively. The SAF lines look more similar straight lines than FSAF, thus the frames are chosen in a segment instead of stretching all over the clip. Better viewed in color.

The input of the algorithm is a series of consecutive frames which extracted from videos. We further expand the method in case of the segment length equals to the whole clip. We call it is Full Selected Active Frames (FSAF) method.

To the SAF method, a video is divided into 16 segments with the same number of frames. In each segment, we apply the above algorithm to collect 16 frames for constructing a 16-frame dataset. For the FSAF method, the most 16 active frames from beginning to the end of a video are selected.

We have a comparison chart of SAF and FSAF as Fig. 1 with three random videos from HMDB51 dataset. It is simple to realize that, the SAF line is little more straightforward than FSAF one. In our implementation, the accuracy of these two methods is similar with a very slight difference. Thus, we only show the SAF results for the representative method.

V. Multi-stream 3D Convolutional Neural Network

Multimodal networks or multi-layer framework are shown promising performance. In our work, we use three streams as shown in Fig. 2. It includes RGB stream, Optical flow stream and, SAF stream.

A. RGB stream

Sequentially, 16 raw RGB images are extracted from one video to form a totally RGB dataset. The images can be resized to the desired resolution. This approach is used for implementing in recent work [12], [13], [24].

B. Optical flow stream

Among many flow estimations such as Lucas-Kanade [29], Brox [30], we employ Farnebäck method [31] develop in [13], [32] to compute optical flow for KTH [33], HMDB51 and UCF101. The Farnebäck optical flow is considered as dense optical flow with fast and low error rate [13] comparing to Lucas-Kanade and Brox methods. Data after being extracted are saved as optical flow images before feeding into Optical flow stream for training. All the training results are saved for predicting process.

C. SAF stream

Based on the method proposed IV, videos are split into 16 segments without overlap and a same number of frames. SAF method is applied to all videos from three datasets and is executed independently with the training process. The output of this stream is also saved and considered as an input of the fusing process.

The multi-stream network architecture is displayed by the Fig. 2. Each stream is similar to the network which describes in Fig. 3. The vote fusion process is also represented in a visual way for being well-understood.

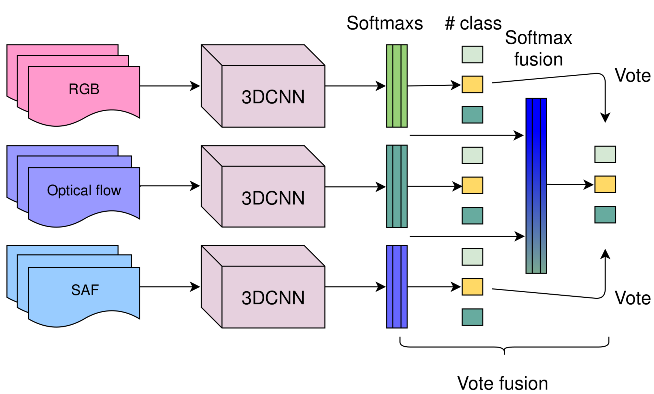

Fig. 2: Multi-stream framework. Three datasets are fed into networks, the outputs as softmax layers are fused to become final softmax. The predict process includes voted classes and score from final softmax layer.

Our network architecture is shown in Fig. 3. The input is a 16-frame block of images which are normalized to 112×112. The first convolution operator has 64 kernels with a size of 3×3. Max pooling is then performed, followed by interleaving convolution/maximization layers (C2-M5). The number of kernels is increased from 64 to 256 to extract more features. The size of a fully connected layer (dense layer) is $512 \times 4 \times 4 = 8192$ which seems to be sufficient to represent the characteristics of human actions. Due to avoid overfitting, we drop out the dense layer with 50 percent. The output is softmax layer which is the probability distribution related to the potential number of classes possible classifications.

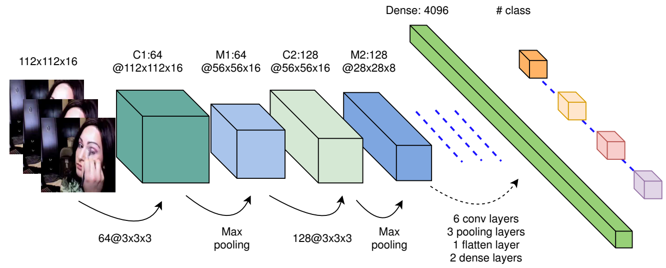

Fig. 3: Network architecture. We use the same structure for all networks, the only difference is the kind of input data including Optical flow images, RGB images and SAF images.

VI. Experiment

A. Datasets

We conduct an evaluation on three popular datasets such as UCF101, HMDB51, and KTH.

First, UCF101 [25] is a large dataset which consists of 101 actions in 13320 videos, every action has 25 groups where each group has a similar background, similar viewpoint, same person and consists 4 to 7 actions (average 131 clips per action). The videos are in .avi format and have dimensions of 320×240 pixels with frame rate is 25 frames per second.

The second dataset is HMDB51 [26], smaller than the first one with 51 actions contained in 6766 clips (average of 132 clips per action). The videos have a spatial resolution of 320×240 and frame rate of 25 fps.

Both two datasets have closely resembled resolution, frame rate and the number of clips per action. However, the classification results are far different (refer to section VII). It is possible that the HMDB51 clips are shorter, the background is more complicated than the UCF101. Beside this, HMDB51 contains several some categories like chewing, talking, laughing, smiling,... and several other categories like drinking, eating. Such similar categories are much difficult in categorizing and easy to be ambiguous.

The last database is KTH [33] which contains six types of human actions including walking, jogging, running, boxing, hand waving and hand clapping. This dataset consists of 600 videos performed by 25 subjects and 4 scenarios. The dimension of the clip is 160×120 and a frame rate of 25 frames per second. The length of each clip is much longer than both two above datasets (approximately 20 seconds per clip).

B. Implementation

We extract RGB images, Optical flow images and SAF images as three datasets. The data constructing process is independent of the whole progress, this causes saving a lot of time. In our experiment, we use a Core i7 computer with 16 cores, 32GB RAM, NVIDIA GeForce GTX 1070 with 8GB memory.

In order to avoid overfitting, the networks are based on the model were pre-trained by Sports-1M dataset which has one million videos in 487 categories [8]. The input data is subtracted by the mean value of sport1M before having fed data to the networks. We apply data augmentations (cropping, flipping, RGB jittering) [1], [34] to increase the diversity of videos. We use Adam optimizer [35] with learning rate 10^{-4} as initiation. The training processes stop after 10000 iterations.

The code is forked from Github [36] with our additional modules for SAF and ensemble process. We share our code at https://github.com/binhhoangtieu/C3D-tensorflow.

VII. EVALUATION

First, we compare the accuracy between the methods which carried on KTH dataset to illustrate the efficiency of SAF method. All these results are conducted on a single stream network. Our network structure applied in this case is described in Fig. 2. The results are shown in Table I. Our network can recognize at a high-level accuracy in almost actions. The best recognizable actions comparing to other methods are **Jogging** and **Running** with 92% and 89.2%, respectively. Secondly, we compare our multi-stream framework with others works. Our networks are shown in 3. Two datasets HMDB51 and UCF101 are used in order to make the comparison between other state-of-the-art achievements. The results in Table II show that: Toward single stream, SAF method achieves best result with 49.4% and 88.1% correspond to HMDB51 and UCF101 dataset. In the manner of using multi-stream, our methods perform improvement on three fusion methods. Among these, three-stream with Vote fusion method gain the best accuracy with 66.2% and 94.9% for HMDB51 and UCF101, respectively.

Method	Boxing	Hand clapping	Hand waving	Jogging	Running	Walking	Average
3DCNN [37]	90	94	97	84	79	97	90.2
Schuldt [33]	97.9	59.7	73.6	60.4	54.9	83.8	71.7
Dollár [38]	93	77	85	57	85	90	81.2
Niebles [39]	98	86	93	53	88	82	83.3
Jhuang [40]	92	98	92	85	87	96	91.7
SAF (Ours)	92	90	91.3	**92**	**89.2**	87.2	90.3

TABLE I: Action recognition for accuracy on dataset KTH. The unit is in percentage. All methods use single-stream structure.

We also implement three fusion methods separately for KTH dataset. The results reported in Table III show that the vote fusion slightly better than Max fusion and Average fusion. Comparing to single-stream networks shown in Table I and Table II, multi-stream structure with vote fusion shows significant improvement in accuracy, increase $3.5\%, 6.8\%, 6.8\%$ comparing to SAF network on KTH, HMDB51, UCF101 respectively.

Input	HMDB51	UCF101	Network structrure
RGB [9]	50	85.2	3-C3Ds
DI [12]	46.8	78.4	C3D
OF [12]	48.9	78.2	C3D
RGB+DI+OF [12]	57.9	88.6	3-C3Ds
RGB [24]	53.1	85.4	3DCNN
DI+RGB+Trajectory [10]	65.2	89.1	2DCNN
SAF	49.4	88.1	C3D
RGB+OF+SAF	64.4	92.9	3-C3Ds+Max fusion
RGB+OF+SAF	65.9	94.2	3-C3Ds+Avg fusion
RGB+OF+SAF	66.2	94.9	3-C3Ds+Vote fusion

TABLE II: Comparision between some state-of-the-art results. The unit is in percentage.

Fusion method	KTH	HMDB51	UCF101
Max fusion	90.8	64.4	92.9
Avg fusion	93.8	65.9	94.2
Vote fusion	93.8	66.2	94.9

TABLE III: Fusion Comparison table

Concretely, we show our deep analysis about the difference between fusion methods. The Table IV makes comparison Max fusion and Average fusion inside Vote fusion method. This happens when all networks recognize absolutely differently. Our framework will apply vote fusion for final labels. In our test case, with KTH dataset, only two instances are unalike and Max fusion truly recognized 1 instance, occupied 50%. With HMDB51, Max fusion correctly predicted 246 instances, occupied 36.66%, Average fusion gained 39.64% in correcting rate. Last, on the UCF101 dataset, the truly recognized rates are 69.65% and 75.55% for Max fusion and Average fusion respectively.

According to these statistics, the max fusion may get a better result in case of applying inside Vote fusion. With this conclusion, in our work, we apply Average fusion for entire Vote fusion progress.

VIII. CONCLUSION

In this paper, we have concretely described an approach on video classification using 3D convolutional networks with Selected Active Frame extracting method. Beside this, we also demonstrate the efficiency in recognizing when applying vote fusion method. Our results show significant improvement comparing the state-of-the-art results.

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Dataset	Test instances	Different labels	Max Fusion		Average fusion	
			Number	Percent	Number	Percent
KTH	129	2	1	50%	0	0.00%
HMDB51	1722	671	246	36.66%	266	39.64%
UCF101	3318	458	319	69.65%	346	75.55%

TABLE IV: Fusion methods comparison. Different labels column represents for a number of labels which are classified differently between streams. The Max Fusion and Average Fusion columns represent for a number of classes which methods are truly recognized. The Percent columns are calculated from the ratio between Number and Different labels columns.

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